

A student team building an autonomous boat from the ground up.

One composite planing hull, two motor pods steering by differential thrust, and a rudder for stability at speed. Built at UW–Madison for a 2-mile course with a 30 pound payload.



4–6 ft
composite hull

30 lb
added payload

2 mi
course

2
motor pods + rudder

OUR APPROACH

01

Hull

Plane well, carry the load, stay light.

- 4–6 ft rigid composite hull with a hybrid planing shape
- One optimization model picks hull geometry, drive, battery and speed together
- Resistance checked in CFD before anything is built

02

Electric

Spend every watt-hour on speed.

- Battery energy storage system within the 55.5 V race limit
- Two motor pods and propellers sized by the model
- Differential thrust steers; a rudder adds stability at speed

03

Autonomous

Drive the course with no one at the helm.

- GPS, IMU and LIDAR sense position, heading and obstacles
- Embedded firmware runs the motors and protects the battery
- Controls software steers the boat around the 2-mile course

TIMELINE

Oct 25 Hull design	Nov 15 Electrical design	Nov 25 Hull built	Jan 25 First drive	Feb 15 First fast drive	Mar 15 Tuning and testing
------------------------------	------------------------------------	-----------------------------	------------------------------	-----------------------------------	-------------------------------------

Sponsor the build

We are looking for funds, parts and skills.

CONTACT

Marley Barrett, Project lead
mhbarrett@wisc.edu